CCIT Report #581
April 2006

# On the Error Exponents of Some Improved Tangential-Sphere Bounds

Moshe Twitto Igal Sason

Technion – Israel Institute of Technology
Haifa 32000, Israel
{tmoshe@tx, sason@ee}.technion.ac.il

March 27, 2006

**Abstract**

The performance of maximum-likelihood (ML) decoded binary linear block codes over the AWGN channel is addressed via the tangential-sphere bound (TSB) and some of its improved variations. The paper focuses on the derivation of the error exponents of these bounds. Although it was previously exemplified that some variations of the TSB suggest an improvement over the TSB for finite-length codes, it is demonstrated in this paper that all of these bounds possess the same error exponent. Their common value is equal to the error exponent of the TSB, where the latter error exponent was previously derived by Poltyrev and Divsalar.